

Option Trading

Volatility Trading and Variance Premium

About me

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Let's start learning!

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Session One Summary

- General option trading philosophy.
- Trading Principles: edge and risk.
- The Efficient Market Hypothesis (EMH).
- Finding edge.
- Inefficiencies and risk premia.

Session Two Overview

- Models and situations.
- Adverse selection and trading costs.
- Why options are appealing to trade.
- Realized volatility.
- Implied volatility.
- The volatility premium.

Finding Trades with Edge

- These are the two types of trades with edge.
- Model driven: Always have a fair value based on a particular model.
- Event driven: Based on special situations, not values.

Examples

- Blackjack: Card counting vs ace tracking
- Sports: Simulations vs over-reaction
- Options: Black Scholes vs variance premium

Pros and Cons of Model-driven

- Model driven needs to have a good model.
Sometimes the model fails but still looks good because the market is pricing in something you aren't.
- Can always be trading.
- Easy to estimate whether a trade is good or bad.
- Often scales well.

Pros and Cons of Model-driven

- The reason we trade options is because volatility is predictable.
- So, we should make an effort to predict it.
- But there comes a point where good enough is enough.
- A simple model, consistently calculated and applied will get you in the right ballpark.
- A complex model will often create more issues than it solves:
 - HF data needs to be cleaned and filtered to remove bid/ask bounce effects.
 - The opening price for many instruments is meaningless.
 - Online estimators need data pipes and computing power.

Pros and Cons of Situation Driven

- If you don't know why a trade exists, it can be hard to know when it will stop.
- Often based on psychological reasons and we can talk ourselves into belief.
- Need to wait for situations to arise.
- Hard to estimate edge and hence size.
- It may not scale.

Embracing Ambiguity

- We should usually look not for situations where we are 100% confident, but instead for cases that we have several *reasons* to believe but no real *proof*.
- Academics look for a t-statistic of 2.
- Traders should look for a t-statistic of around 1, together with other types of evidence.
- If a trade is totally proven, it will soon cease to exist.

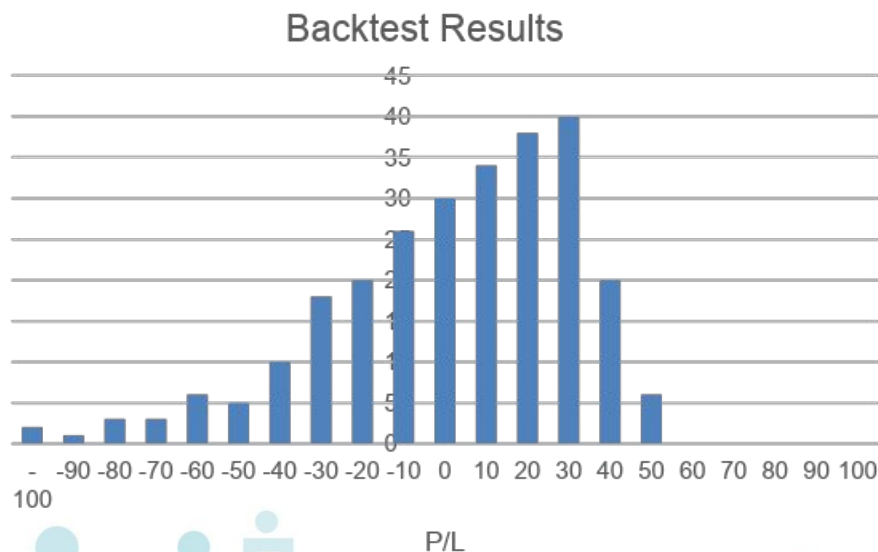
Becoming Comfortable with Uncertainty

- It can be anxiety inducing to trade like this. These thoughts may help (or not, I don't know you and I'm not a psychiatrist).
- Use “mental accounting”.
- Tell yourself you've already lost the money, then any other result is upside.
- Remember that in ten years, the bad day you are having won't even be one you remember.
- Even if your idea is bad, remember it is as hard to negative edge as positive edge, so you are most likely just losing costs, not finding negative alpha.

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Adverse Selection

- The biggest problem in trading is that you might be the dumb one.
- The counterparty is doing the trade for a reason as well.

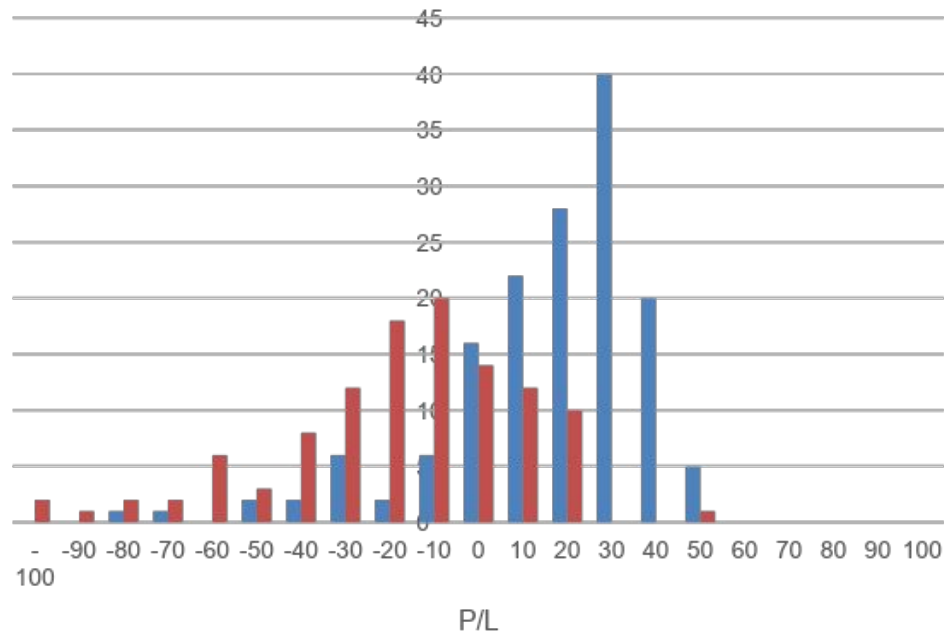


Adverse Selection

- It isn't as simple as saying "we have positive EV so we can just do all the trades and ride out the variance involved".
- We have implicitly assumed all of our trades are from the same true distribution.
- More likely that there is a "good" set of trades and a "bad" set of trades.

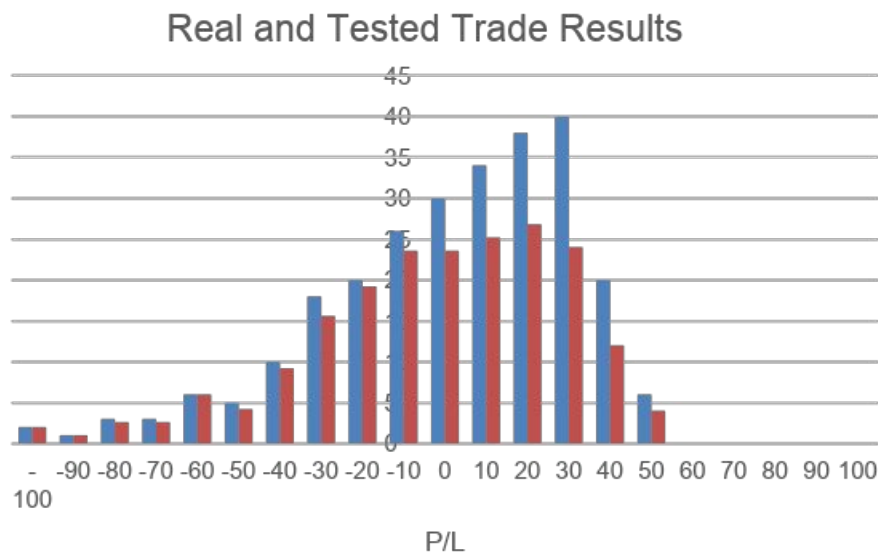
Adverse Selection

"Good" and "Bad" Trades



Adverse Selection

- In reality, we are much more likely to get the “bad” trades.
- The “good” trades are much more competitive.



Transaction Costs

- Transaction costs include: the bid/ask spread, clearing fees, exchange fees and brokerage fees.
- Every time you trade there are costs.
- Insidious, because each cost is small. But they add up.
- It is easily possible to pay 20% of expected profit in fees.
- They have an infinite, negative Sharpe ratio.
- The best way (by far) to improve a winning strategy is to reduce costs.

Adverse Selection and Costs

- To avoid adverse selection, we should take all trades.
- To reduce costs, we should try to not cross bid/ask spread.
- These are contradictory (no-one said this this was easy).
- Be aggressive when entering and conservative when rebalancing or hedging.
- Avoid things with wide spreads or low volumes.
- Look for lower rates and fees.

Options and Trading

- General rule: simplest products trade in the most complicated ways.
- Futures: market makers use many “tricks” such as book stacking, spoofing, flipping and stop hunting.
- Vanilla options: capturing edge is easy and traders are differentiated by risk management.
- Exotic Options: edge comes from structuring, legal and sales.

Why Options?

- The main point of this course is that options depend on *volatility*, the fluctuations of the market.
- The market going up or down has no bearing on the instantaneous **price** of an option.
- Because the major driver of option prices is volatility, trading options is implicitly trading volatility.

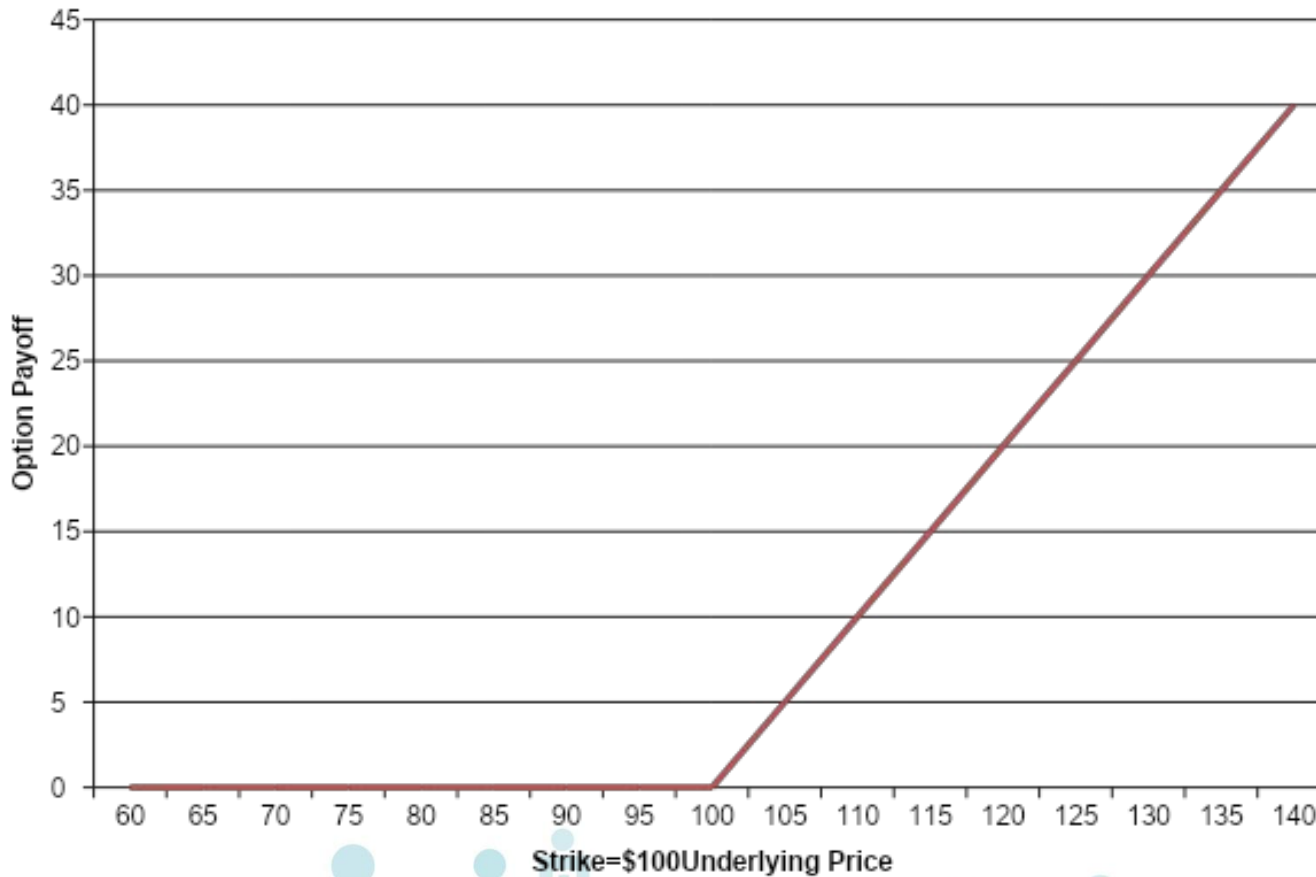
Why Options?

“For predicting future values, historic data appear to be quite useful with respect to standard deviations... and virtually useless for expected returns.” -- William Sharpe

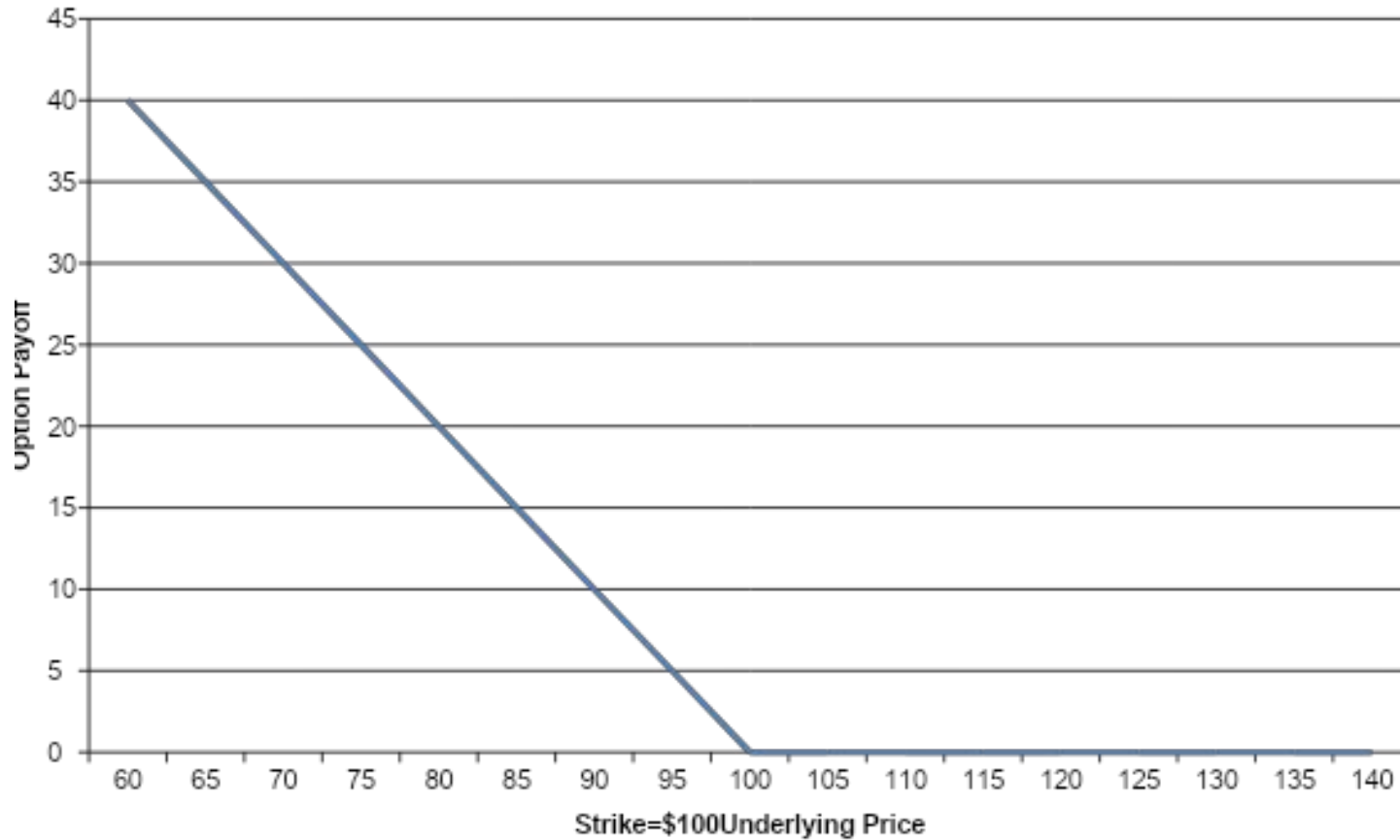
Option Payoff Functions

- The payoff is the value of the option at expiration.
- If the underlying price is S and the strike is X :
- Call payoff is $\text{MAX}(S-X, 0)$
 - So, the holder makes money if the stock goes above the strike.
- Put payoff is $\text{MAX}(X-S, 0)$
 - So, the holder makes money if the stock goes below the strike.

Payoff Diagram: A Call

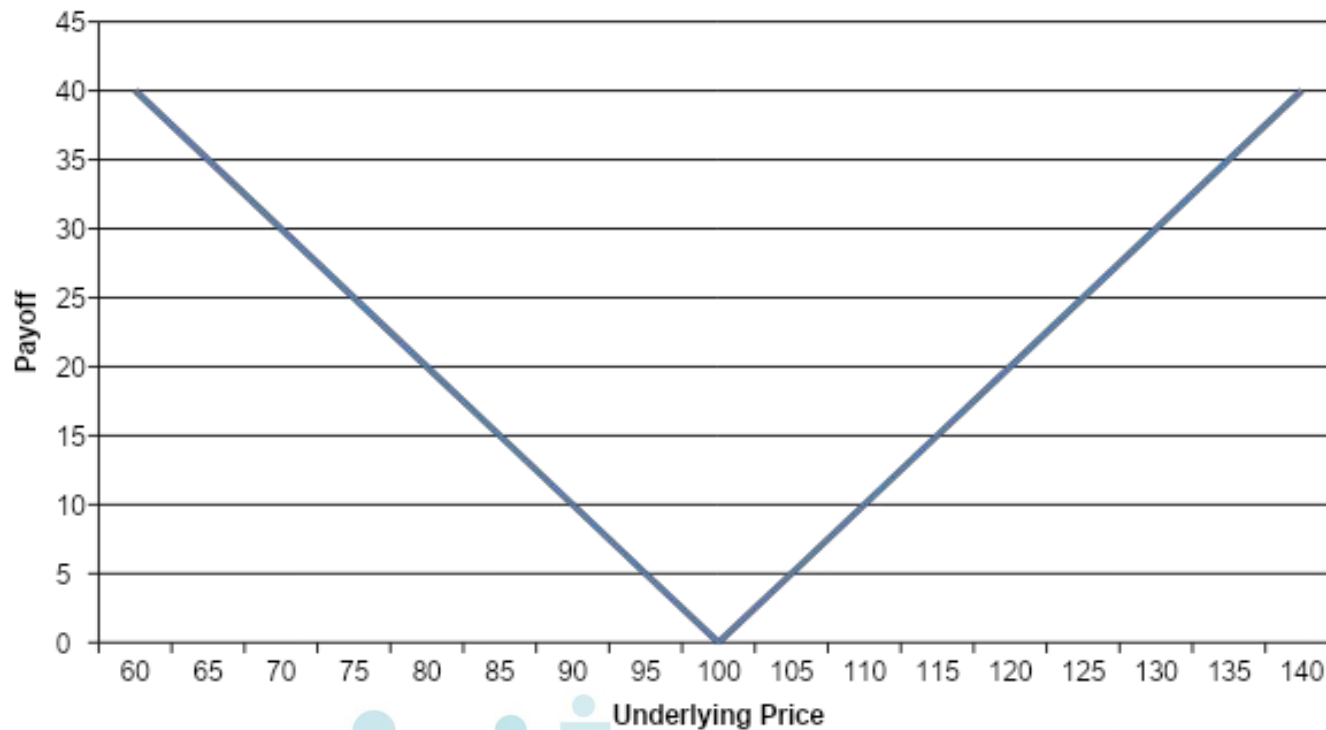


Payoff Diagram: A Put



Example: Straddle

- A long straddle is made by buying a put and a call with the same strike and expiration.



Example: Straddle

- A straddle is a bet on the mean absolute deviation (MAD) of the underlying.
- Standard deviation is a re-scaled MAD but is easier to work with.
- So, a straddle is a standard deviation, or volatility, bet.
- Further, because option directional exposure can be removed, their price CAN'T depend on directional exposure.
- You don't get paid for risk you don't have to take.

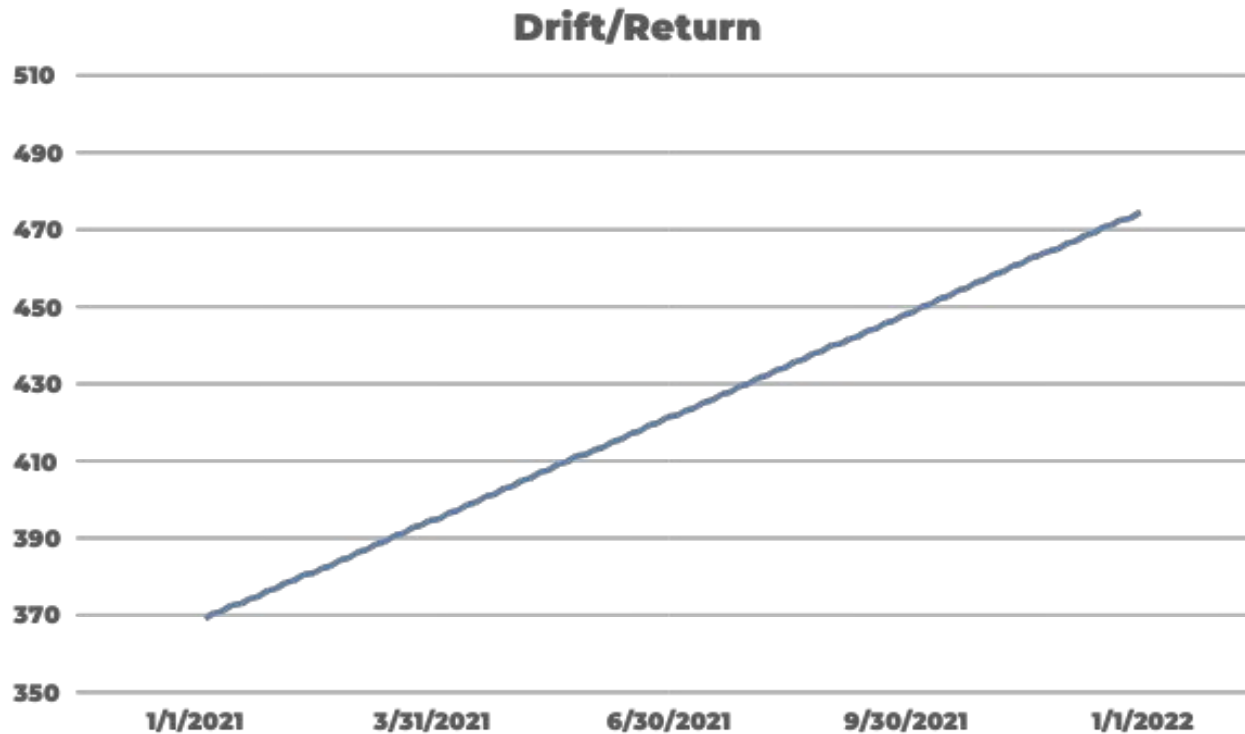
What is Volatility?



What is Volatility?

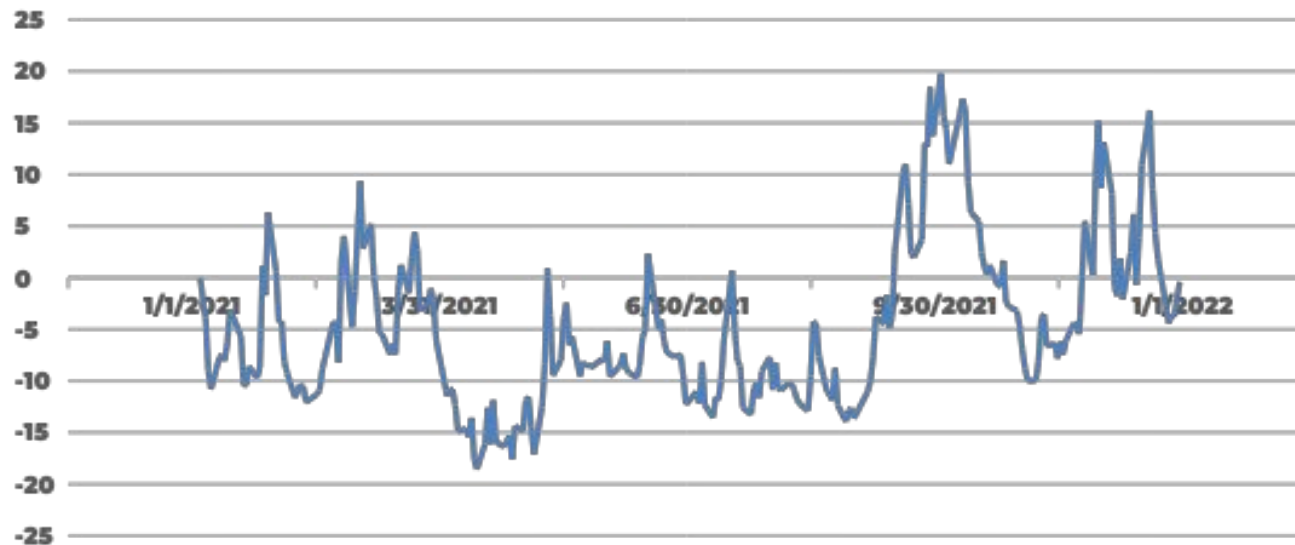
- A time series has two components.
- A trend (or drift): In 2021, SPY rose 27%.
- But there was considerable variability around this return.
- Informally, volatility is a way to quantify these fluctuations.

What is Volatility?



What is Volatility?

"Wiggles"

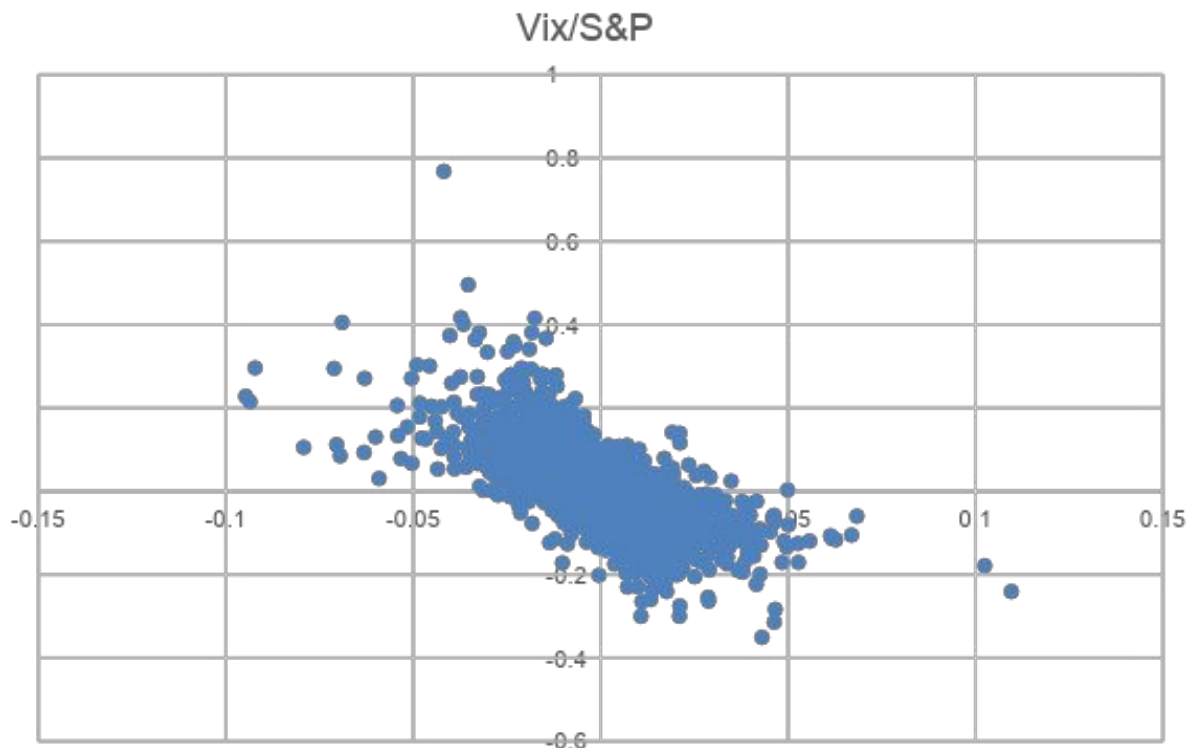


Facts about Volatility

- Negatively correlated to return.
- Mean-reverting.
- Generally, much larger than returns e.g., SPY in 2021 had a daily return of 0.1% but a daily volatility of 0.8%.

Correlation to Returns

- S&P 500 volatility/return correlation is -0.70 (weekly is -0.71 , monthly is -0.68)



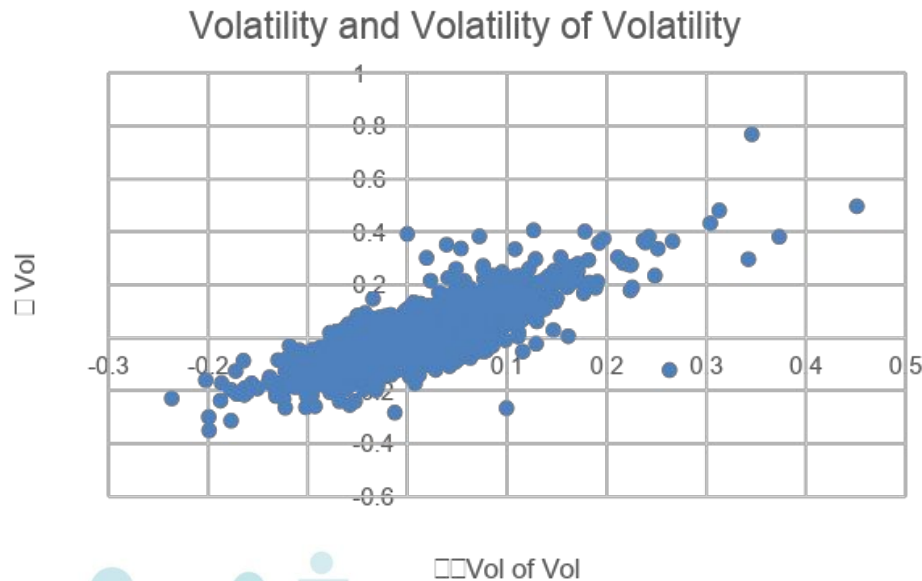
Correlation to Returns

- This effect is less prevalent for commodities and individual stocks.
- Over the last five years the correlations have been.

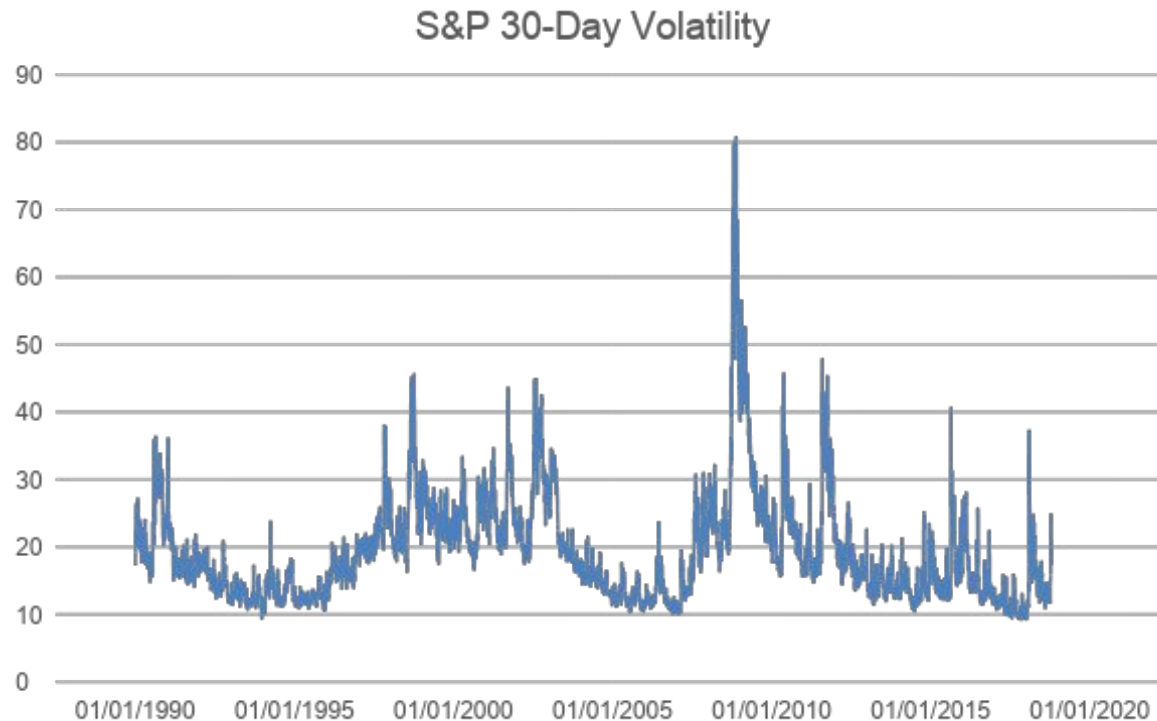
Product	Correlation
Oil	-0.47
Gold	0.15
AAPL	-0.59
AMZN	-0.50
IBM	-0.05

Vol and Vol of Vol

- Can see from the picture that volatility is itself more volatile at high levels.
- Correlation is 0.8.

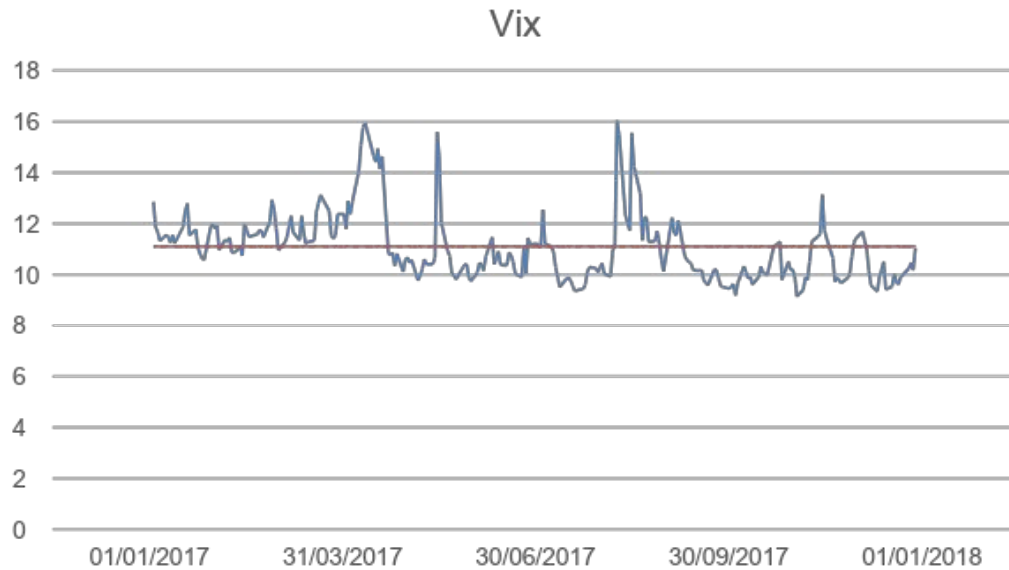


Reversion



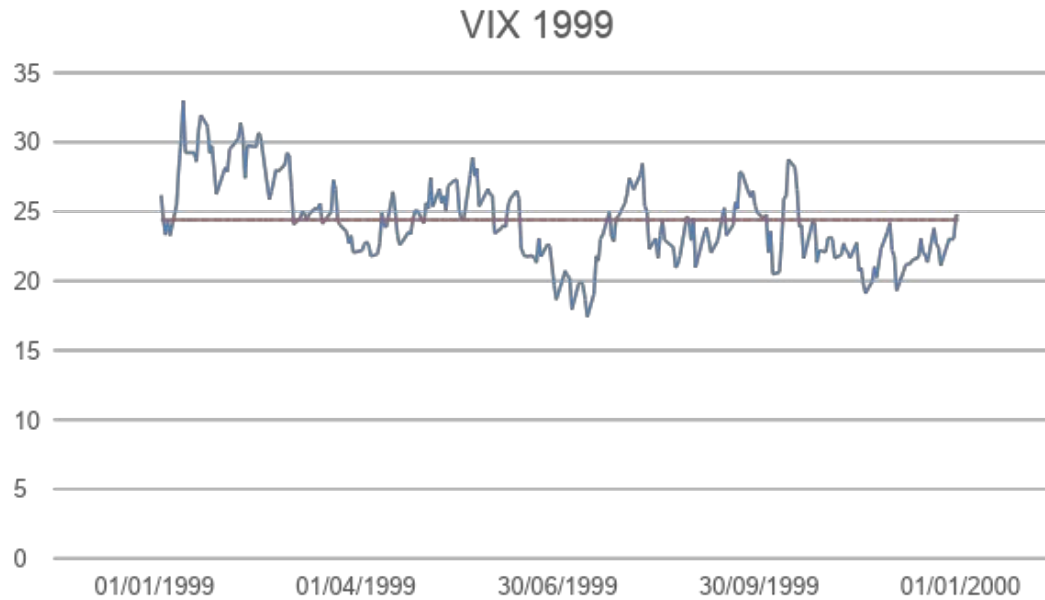
- Doesn't really “go anywhere”.
- Mean reverts to time varying mean.

Volatility in 2017



- Mean: 11.1

VIX 1999



- Mean: 24.4

Reversion

- VIX since 1990:
- Daily autocorrelation= -8.9% (51% of days have opposite sign return of previous day)
- Weekly autocorrelation= -19.8% (56% reversal)
- Monthly autocorrelation= -16.6% (60% reversal)

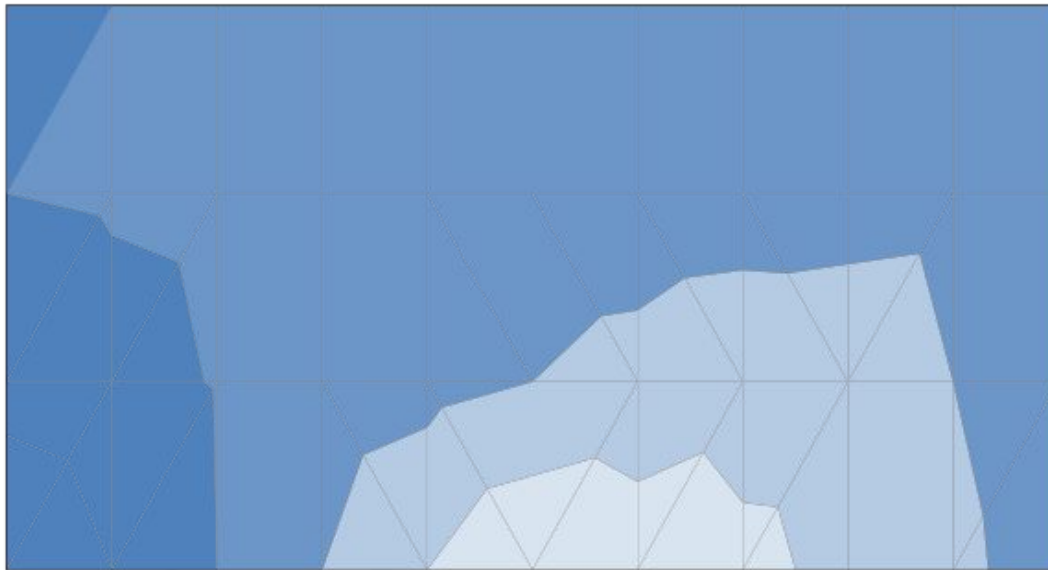
Two (very different) Volatilities

- Historical volatility: Calculated from the returns of the underlying instrument.
- Implied volatility: Option market's guess of future volatility. Specifically, we get the option price from the market and calculate what volatility we need to put into the BSM model to get that number.
- Historical => Backward looking
- Implied => Forward looking

Implied Volatility

- There is no single implied volatility.
- Generally, every strike and maturity pair has its own implied volatility (forming the IV surface).

IV Surface



- Volatility as a function of strike is called “smile” or “skew”.
- Volatility as a function of time is called “term structure”.

Reasons for Smile

- Mathematically: Model misspecification, the BSM model isn't an accurate representation of reality.
- Financially:
 1. Correlation between underlying and volatility: as the stock drops, implied volatility increases so lower strikes have higher implied volatilities.
 2. Leverage effect: as stock price drops, a company becomes more leveraged as their debt/equity ratio increases.
 3. Longshot bias: far OTM options are cheap in dollar terms, so they are bought as lottery tickets. This pushes the volatility up.

Reasons for Term Structure

- Mean reversion: If the market is currently volatile, short dated options will have high implied vols, but the back months will increase far less as we expect the effect to be temporary.
- Different things happen in different time periods:
 1. Earnings announcements
 2. Takeover speculation
 3. Different crops
 4. Different weather patterns

Volatility Trading

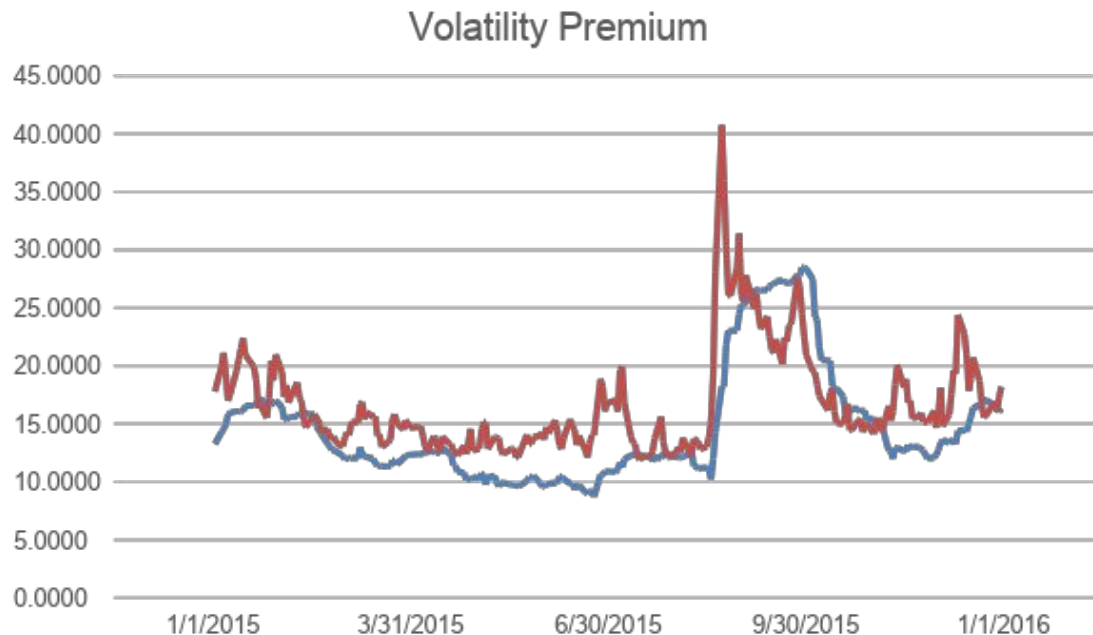
- The implied volatility is like the bookie's odds on the game, the price of the bet that gets locked in.
- Then the game is played, or the stock realizes a volatility.
- If implied and realized volatilities are different, the option was mispriced.
- Unless you have this volatility edge, you are making bad bets.

Volatility Trading

- So next we need to find situations where we expect realized volatility to be different to the implied volatility.
- Model driven methods are based on time series analysis, using the empirical regularities documented earlier.
- Situational methods can be based on unusual events such as elections or earnings, or unusual times such as weekends.
- But we must also acknowledge the variance (or volatility) premium.

Volatility/Variance Premium

- Implied volatility > subsequent realized volatility.
- S&P shown below.

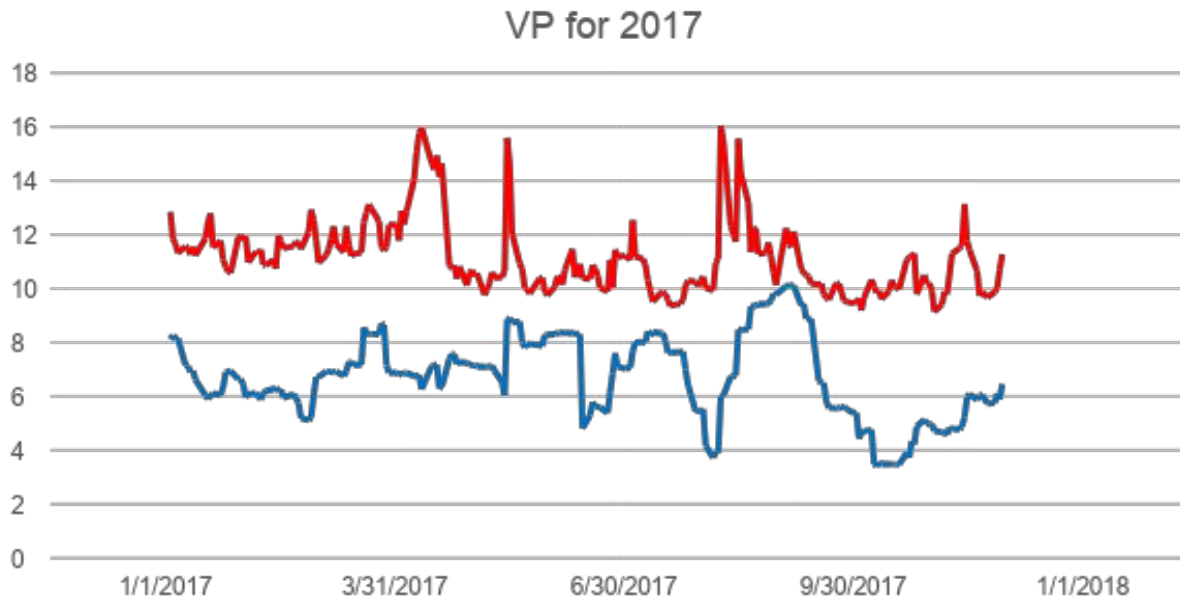


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- Average premium=3.6 volatility points.
- Median premium=4.0 volatility points.

Volatility Premium

- Persistent even at very low volatility levels.
- 2017 shown below.



- Average premium=4.4 volatility points.
- (about 40% of the implied volatility level).

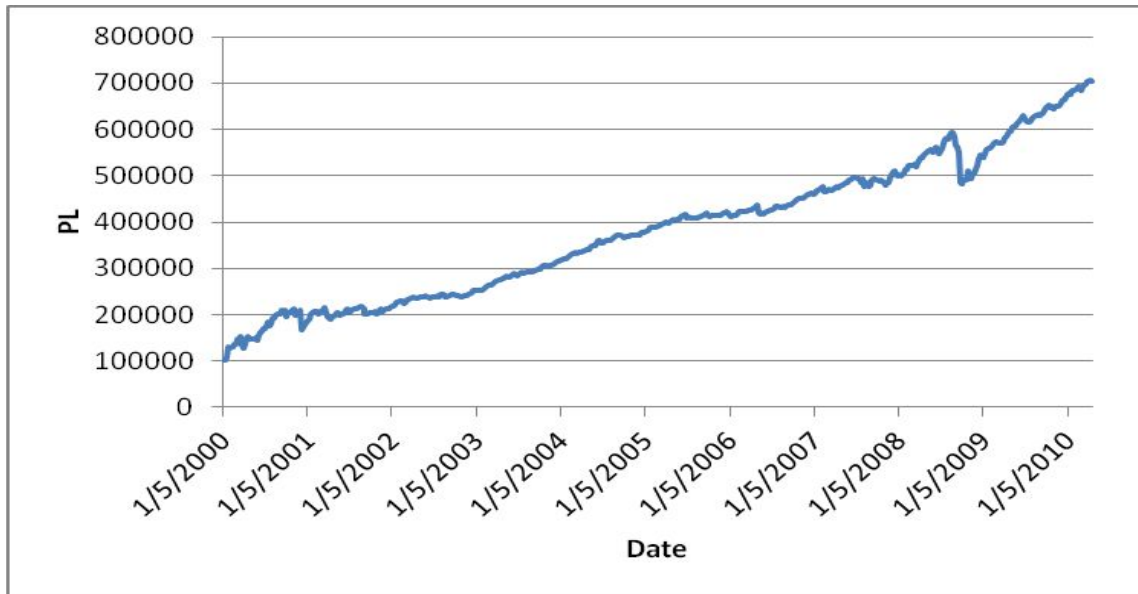
Volatility Premium

- As a percentage, the premium is generally *higher* at low volatility levels.
- From 1990 to 2018:

VIX Level	Volatility Premium (as %)
<15	0.27
15<20	0.26
20<25	0.20
25<30	0.20
30<35	0.18
35<40	0.14
>45	0.10

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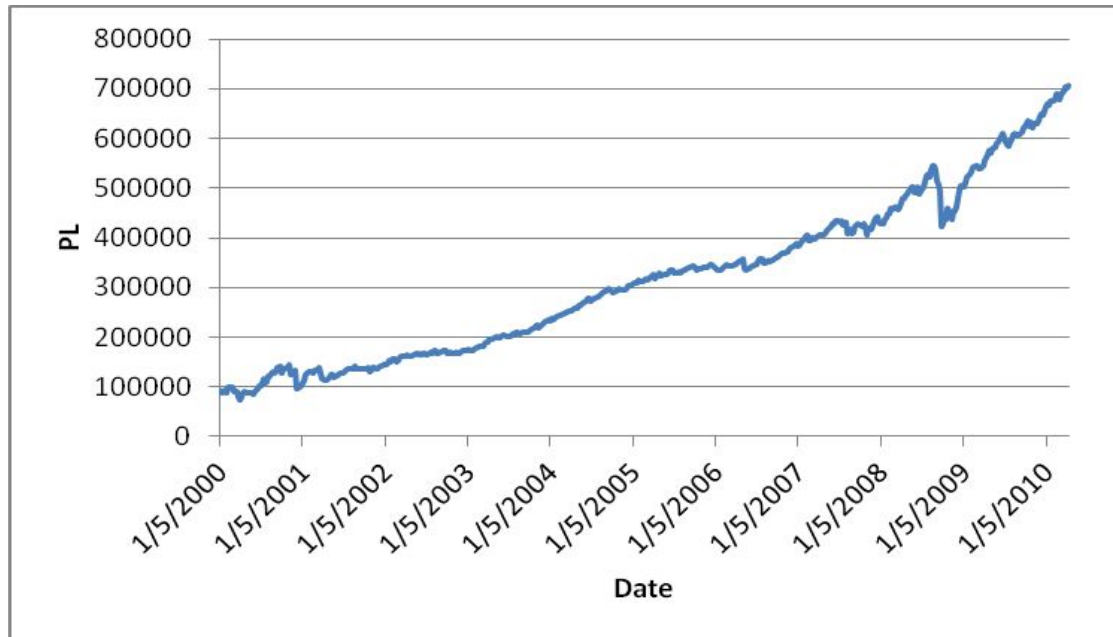
Volatility Premium



- Selling 10 delta QQQ, 2nd month strangles (using account value for margin).

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Volatility Premium



- Selling ATM QQQ, 2nd month straddles.
- (Same vega exposure as with strangles).

Why Does the Premium Exist? - Insurance Premium

- Both calls and puts provide insurance.
 - Puts against a crash hurting an existing portfolio.
 - Calls against FOMO (Fear Of Missing Out).
- All insurers charge a premium for their products so they can make a profit.
- This is no different from any other shop.

Why Does the Premium Exist? - Fear of Atypical Events

- People vastly over-estimate the danger of extreme events.
- Terrorism kills 180 Americans a year (about 8 if we exclude 9/11).
- Heart disease kills 600,000 Americans each year.
- This mistake means options are overpriced.
- “Black Swans” don’t happen very often.

Volatility Premium in General

- Exists in most products, but usually lower than indices.
- Over the last decade some premia have been:

Product	Average RV (%)	Average VP (%)
Oil	31.9	5.9
Gold	14.4	2.7
AAPL	26.2	4.1
AMZN	31.0	3.1
GOOG	22.6	2.6
GS	25.5	4.3
IBM	19.6	4.0

Psychology

- Psychological traits like discipline, control and perseverance are not the primary drivers of success.
- Trading is a performance-based activity and, like any other, psychological attributes are not sufficient.
- No matter how disciplined you are, you won't be playing cricket for India.
- Conversely, many successful traders are lunatics.

Psychology

- Trading psychology is a set of tools that “keep you in the game”.
- It can’t give you edge but can destroy your edge.
- Emotional version of risk management.

Common Issues

- Traders' block, being unable to do a trade.
- Going on tilt, wildly over-trading.
- Staying in a position when it has no edge.
- Being unable to see why you could be wrong.

Cognitive Behavioral Therapy

- Doesn't attempt to solve the root causes.
- Gives a set of tools and tricks to cope with the problems.
- Self coaching.
- Used for addiction and depression treatment.

Observe Yourself

- Try to observe yourself as you would someone else.
- Need to know your bad habits before you can fix them.
- Seeing the problems as “someone else's” put them in perspective.
- Talk to yourself as if you were your friend.

Externalize Thoughts

- Writing thoughts down helps put them in perspective.
- “I suck” is a powerful thought that looks stupid on a piece of paper.
- Next, write a rebuttal.
- Documenting results gives real evidence that you don’t suck.

Talk



- The way we talk to ourselves is very powerful.
- Invent an imaginary friend...
- Or get real friends...

Learn Actively

- Reviewing trades, analysis and process is vital to improve.
- But also helps psychologically.
- If your “mistakes” are re-framed as “learning opportunities” then they can be seen as good things.

Set Realistic Goals

- Goals are important.
- Unrealistic goals will never be met, so the result is that you will always see yourself as a failure.
- Try not to compare yourself to others.

“Comparison is the thief of joy.”
-President Teddy Roosevelt.

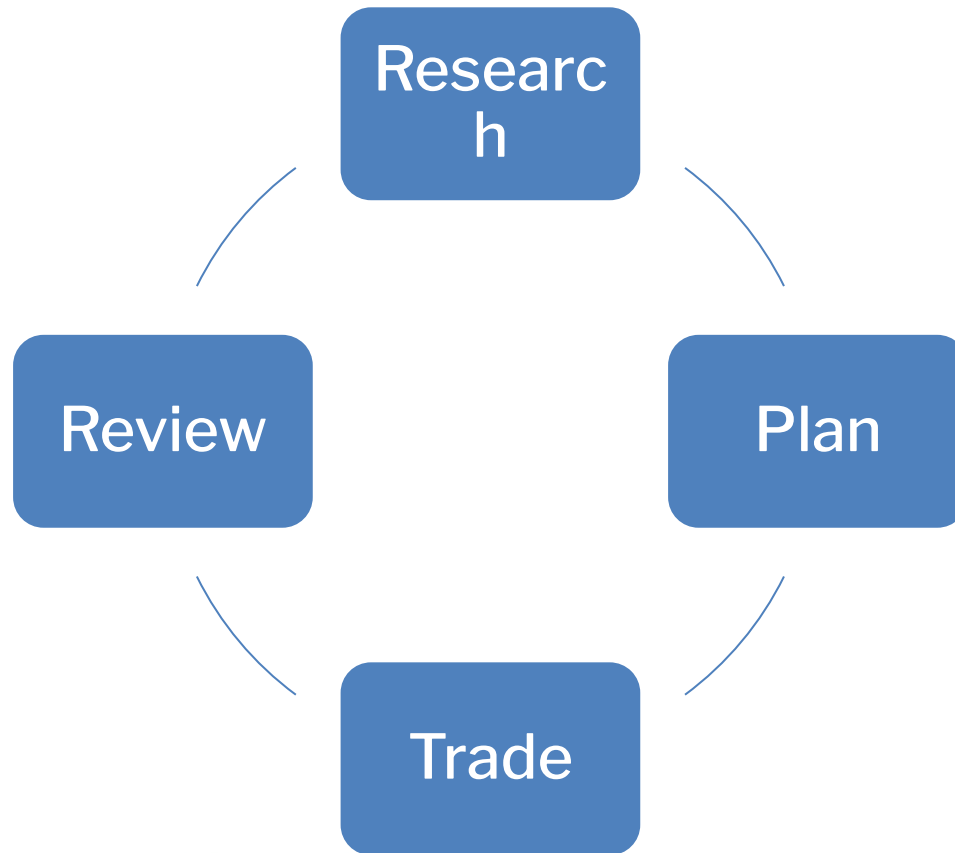
Get a Life

- Making trading the basis for your self esteem is a terrible idea.
- Trading has massive short-term randomness.
- Obsession is unlikely to help.
- Don't "watch" the market. At best it is a waste of time, but usually it leads to emotional over involvement.

Things that Don't Work

- Trying harder. Doing more of the same thing won't lead to different results.
- “Finding your own style”. You need to be comfortable with a real edge, not base trades on what makes you comfortable.

Process



Conclusion



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